

Elias D. Crum

COMPUTER SCIENCE ENGINEERING PHD CANDIDATE · BIOINFORMATICIAN

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Profile

Computer science engineering PhD candidate and bioinformatician working on interoperable, privacy-aware clinical genomic data. Research spans RDF knowledge graphs, ontology definition, federated SPARQL querying, and Solid-based data sharing, with prior experience in microbial genomics and direct clinical care support.

Technical Skills

Semantic Web: SPARQL, RDF serialization, ontology definition, Solid

Programming/Querying Languages: Python, BASH, R, Java, JavaScript, TypeScript, SPARQL, SHACL, HTML, CSS, Vue, LaTeX

Research tooling: GitHub Actions, NGINX, Conda, Docker, Cytoscape

Languages: English (native), Dutch (A1), French (A2)

Research & Clinical Experience

Graduate Researcher, Ghent University & VITO NV

Ghent, Belgium

Ph.D. Candidate

Oct. 2023 - Present

- Developing framework for clinical genomic data storage, sharing, and querying.
- Implementation of the framework utilizes the Solid protocol.
- Experience with SPARQL, RDF, ODRL, Linked Data, decentralized querying, and ontology definition.

SIB Swiss Institute of Bioinformatics

Lausanne, CH

Research Stay

Mar. - Apr. 2025 (2 months)

- Knowledge Representation Unit (KRU): public SPARQL endpoint maintenance and Solid Cockpit development for federated querying across endpoints and research-data Solid pods.
- FWO Long Research Stay V416635N; CHIST-ERA TRIPLE (CHIST-ERA-22-ORD-09).

Master's Research, Loyola University Chicago (Putonti Lab)

Chicago, IL, USA

Master's Researcher

Aug. 2020 - Aug. 2022

- Identified and characterized novel, medically relevant bacteria and bacteriophage sequences.
- Experience with gene annotation and clustering, taxonomic classification screening, gene-similarity network production, genetic engineering and phage virulence augmentation.

Emergency Department, University of Chicago Medical Center

Chicago, IL, USA

Medical Scribe

Nov. 2021 - Jan. 2023

- Directly assisted physician care of patients through patient intake charting, ordering tests, and other EPIC platform mediated physician logistical support.
- Operated EPIC software for medical documentation.

Education

Ghent University

Ghent, Belgium

Doctor of Computer Science Engineering (in progress)

Oct. 2023 - Present

KNowledge on Web Scale (IDLab, UGent) / Digital Precision Health (VITO NV)

Loyola University Chicago

Chicago, IL, USA

Master of Science (Bioinformatics)

Aug. 2020 - Aug. 2022

Graduated with Bioinformatics Honors · GPA: 4.00 / 4.00

Loyola University Chicago

Chicago, IL, USA

Bachelor of Science

Aug. 2017 - May 2021

Double Major: Bioinformatics & Biology · Minor: History · GPA: 3.991 / 4.000

Selected Software & Semantic Resources

VCF-RDFizer (github.com/ecrum19/VCF-RDFizer)

VCF-to-RDF conversion with RML/RMLStreamer, a dedicated vocabulary, and compression options.

Active · updated Sept 2026 · Papers: SWAT4HCLS 2026; BioHackrXiv 2025; SWAT4HCLS 2025

Solid Cockpit (github.com/KnowledgeOnWebScale/solid-cockpit)

Browser interface for Solid Pod data, access permissions, notifications, and SPARQL querying.

Active · updated Aug 2026 · Papers: ESWC 2026

Ontology Companion Generator (github.com/ecrum19/ocg)

Configurable npm generator for ontology documentation, exploration, and visualization websites.

Active · updated Sept 2026

VCF-Core Vocabulary (github.com/ecrum19/vcf-core-vocabulary)

Converter-independent vocabulary and SHACL shapes for representing VCF 4.5 genomic variant data in RDF.

Active · updated Sept 2026 · Papers: SWAT4HCLS 2026; BioHackrXiv 2025; SWAT4HCLS 2025

Selected Publications

Does SPARQL federation work in the real world? A case study over large biological SPARQL endpoints

E. Crum, B.-E. Tam, J. Hanski, A.-C. Sima, T. Mendes de Farias, J. Bolleman, R. Taelman

International Semantic Web Conference (ISWC) 2026 · 2026 · Conference Paper · Accepted

Genomic sequence data sharing for clinical practice: A scoping review

E. Crum, R. Taelman, B. Buelens, G. Ertaylan

Computers in Biology and Medicine · 2026 · Journal Article

PErsoNal Genome QUery IN healthcare and clinical practice (PENGQUIN)

E. Crum

European Semantic Web Conference (ESWC) 2024 · 2024 · PhD Symposium Paper

Coliphages of the human urinary microbiota

E. Crum, Z. Merchant, A. Ene, T. Miller-Ensminger, G. Johnson, A. J. Wolfe, C. Putonti

PLoS ONE 18(4), e0283930 · 2023 · Journal Article

Fellowship & Selected Awards

Strategic Basic PhD Fellowship Awardee (2024-28) · The Research Foundation - Flanders (FWO)

Four-year fellowship supporting privacy-aware clinical genomic data management and querying.

Albare Award (2026) · SWAT4HCLS 2026

Poster award for VCF-to-RDF research, selected by consensus ranking across generative-AI models.

Best Paper (Group) (2024) · International Semantic Web Summer School 2024

Best scientific report award for the ISWS 2024 group research project.

Best Poster (Individual) (2024) · International Semantic Web Summer School 2024

Best poster award for presenting the PENGQUIN clinical genomics research framework.

Teaching & Communication

Department of Biology, Loyola University Chicago

Adjunct Professor of Biology

- Taught two sections of Biology Lab I and II each semester, with 24 students per section.

Chicago, IL, USA

Aug. 2021 - May 2023

For full interactive publication list, project links, talks, and a complete CV please visit my website:

<https://ecrum19.github.io/eliascrum/about/cv>